# E-Commerce: Business, Technology, Society

Executive Midterm Study Guide & Core Knowledge Architecture

Course: E-Commerce | Scope: Lec 1, Ch 2, Ch 4, Ch 5 | Business Models | Security & PKI | Online Payments

## 01 Introduction to Electronic Commerce & Fundamentals (Lecture 1)

Electronic Commerce (EC) represents digitally enabled commercial transactions between and among organizations and individuals. It transforms traditional business through telecommunications and the Internet.

### 1.1 Diverse Perspectives of E-Commerce

| Perspective | Core Definition & Scope |
|---|---|
| **Commercial (Trading)** | Provides the capability of buying, selling, and exchanging products, services, and information over computer networks. |
| **Business Process** | Doing business electronically by completing workflows over networks, substituting digital information for physical processes. |
| **Service** | A tool that addresses the desire of firms, consumers, and management to cut service costs while improving the quality of goods and increasing delivery speed. |
| **Collaborative & Learning** | Enables online collaborative work, inter-organizational learning, information sharing, and global team coordination. |

### 1.2 Major Classifications by Market Relationship

- **B2C (Business-to-Consumer):** Online businesses selling merchandise or services directly to individual consumers (e.g. Amazon, Walmart.com). Highly visible to the public.
- **B2B (Business-to-Business):** Businesses selling to other businesses (e.g. Grainger, Boeing parts procurement). **Accounts for the vast majority of overall e-commerce dollar volume**.
- **C2C (Consumer-to-Consumer):** Consumers selling directly to other consumers, facilitated by a platform (e.g. eBay auctions, Craigslist, Etsy).
- **C2B (Consumer-to-Business):** Consumers setting prices or requirements and businesses competing to satisfy them (e.g. Priceline's "Name Your Own Price", reverse auctions).

- **Intra-business EC (B2E - Business-to-Employee):** Internal corporate portal activities, training, benefits selection, and employee self-service on private intranets.
- **Non-business EC:** Religious, academic, non-profit institutions, and social agencies using the Internet to reduce costs and accept contributions.
- **B2G & G2C:** Business-to-Government (public e-procurement) and Government-to-Citizen (online tax filing, renewing driver's licenses).

### 1.3 Pure vs. Partial E-Commerce (The 3 Dimensions)

EC is classified along 3 dimensions: **(1) The Product/Service**, **(2) The Business Process**, and **(3) The Delivery Agent/Intermediary**.

- **Traditional Brick-and-Mortar:** All 3 dimensions are physical (physical store, cash payment, physical product).
- **Partial E-Commerce:** At least one dimension is digital while others are physical. Example: buying a physical paperback book online on Amazon and having FedEx deliver it.
- **Pure E-Commerce:** All 3 dimensions are entirely digital. Example: buying and downloading an ebook, software license, or streaming music directly online.

### 1.4 Benefits and Limitations of E-Commerce

| Dimension | Key Benefits | Limitations & Barriers |
|---|---|---|
| **For Organizations** | Global market reach, significant cost reductions, 24/7 continuous operations, rapid time to market, mass customization, and supply chain disintermediation. | **Technical:** Bandwidth constraints, software integration with legacy ERP systems, evolving security standards. |
| **For Consumers** | Unmatched convenience, massive product selection, instantaneous price comparison, customer reviews, and immediate delivery of digital goods. | **Non-Technical:** Lack of trust, privacy and security fears, inability to physically inspect items before purchase, resistance to change. |
| **For Society** | Enables telecommuting (reducing traffic congestion and pollution), delivers education/healthcare to remote rural communities. | Digital divide, potential loss of local community retail jobs. |

# 02 E-Commerce Business Models & Concepts (Chapter 2)

**Definition of a Business Model**

A **Business Model** is a set of planned activities designed to result in a profit in a marketplace. A business plan documents this model, and an e-commerce business model leverages Internet unique qualities.

## 2.1 The 8 Key Elements of a Business Model

| # | Element | Key Question Addressed | Core Concept & Examples |
|---|---|---|---|
| 1 | **Value Proposition** | Why should the customer buy from you? | Defines how a product satisfies customer needs (convenience, lower price, personalization). |
| 2 | **Revenue Model** | How will you earn money and generate profits? | Advertising, Subscription, Transaction Fee, Sales, Affiliate models. |
| 3 | **Market Opportunity** | What marketspace do you intend to serve, and what is its size? | Realistic financial potential in the designated target niche. |
| 4 | **Competitive Environment** | Who else occupies your intended marketspace? | Direct competitors (similar products) and indirect competitors (substitute products). |
| 5 | **Competitive Advantage** | What special advantages does your firm possess? | First-mover advantages, brand prestige, lower cost structures, patents, network effects. |
| 6 | **Market Strategy** | How do you plan to promote your products to attract customers? | Pricing, promotion channels, social media campaigns, SEO, distribution. |
| 7 | **Organizational Development** | What types of organizational structures are necessary? | Department divisions, reporting hierarchies, and functional workflow allocation. |
| 8 | **Management Team** | What experience and background are important for leaders? | Crucial for venture capital; demonstrates operational competence and execution ability. |

## 2.2 Major B2C Business Models

- **Portal:** Offers search tools and integrated content packages. *Horizontal Portals* (Yahoo, MSN) cover diverse topics. *Vertical Portals / Vortals* (Sailnet, WebMD) focus on niche industries.
- **E-tailer:** Online retail store. Includes *Virtual Merchants* (pure-play like Amazon/Wayfair), *Clicks-and-Mortar* (multi-channel like Walmart, Best Buy), *Catalog Merchants* (L.L. Bean), and *Manufacturer-Direct* (Dell, Apple).
- **Content Provider:** Distributes digital content (news, music, video). Monetizes via subscriptions (Netflix), micropayments, syndication, and advertising (WSJ, CNN).
- **Transaction Broker:** Processes online transactions (Expedia, E*Trade). Value proposition is saving consumers time and money through instant execution.

- **Market Creator:** Uses Internet technology to create digital platforms where buyers and sellers meet, negotiate, and transact (eBay, Uber, Airbnb).
- **Service Provider:** Sells online services (Google Maps, Carbonite cloud backups, Dropbox).
- **Community Provider:** Creates digital environments where people with similar interests congregate, share content, and communicate (Facebook, LinkedIn, iVillage).

### 2.3 Major B2B Business Models

- **E-distributor:** Supplies indirect MRO goods (maintenance, repair, and operations) directly to thousands of business buyers (e.g. W.W. Grainger).
- **E-procurement:** Digital markets creating and selling access to digital procurement software (e.g. Ariba).
- **Exchanges (Net Marketplaces):** Independently owned digital marketplaces connecting thousands of commercial suppliers and buyers for spot purchasing.
- **Industry Consortia:** Industry-owned vertical markets owned by a small group of dominant purchasing enterprises.
- **Private Industrial Networks (PTX):** Digital networks coordinating supply chain communications and collaborative design between a dominant firm (e.g. Walmart) and its key tier-1 suppliers.

## 03 Building an E-Commerce Presence: Websites, Mobile Sites, and Apps (Chapter 4)

Developing an e-commerce presence requires synthesizing business objectives, technical infrastructure, software engineering, and mobile responsiveness.

### 3.1 The Systems Development Life Cycle (SDLC) - 5 Steps

| Step | Phase Name | Primary Activities & Deliverables |
|---|---|---|
| 1 | **Systems Analysis / Planning** | Identify **Business Objectives** (what the site should do), **System Functionalities** (features needed), and **Information Requirements** (data elements required). |
| 2 | **Systems Design** | **Logical Design:** Data flow diagrams (DFD), entity-relationship models, processing logic.<br>**Physical Design:** Hardware servers, software platforms, database engines, and telecommunications specs. |
| 3 | **Building the System** | Decide between *In-house* build vs. *Outsourcing* development; choose hosting architecture (Self-hosting, Co-location, or Cloud/Managed Hosting). |
| 4 | **Testing** | **Unit Testing:** Testing individual program modules.<br>**System Testing:** Testing the entire integrated multi-tier system under load.<br>**Acceptance Testing:** Verifying system satisfies client business requirements. |
| 5 | **Implementation & Maintenance** | 24/7/365 site monitoring, continuous security patching, benchmarking, database reorganization, and ongoing feature updates. *Dominates long-term lifetime costs*. |

### 3.2 Multi-Tier Architecture & Performance Optimization

- **Multi-Tier Hierarchy:**
  - **Tier 1 (Client):** Browser or mobile app rendering HTML/CSS/JS.
  - **Tier 2 (Web Server):** Serves static assets, handles SSL/TLS termination, routes dynamic queries (e.g. Apache, NGINX).
  - **Tier 3 (Application Server):** Processes core business logic, shopping carts, taxes, and inventory workflows.
  - **Tier 4 (Database Server):** Manages relational/ACID data persistence, indexes, and transactions.
- **Optimization Strategies:** Caching in memory (Redis, Memcached), Content Delivery Networks (CDNs like Cloudflare), database connection pooling, image compression, and minimizing HTTP round-trip requests.

### 3.3 Mobile Presence: Responsive Web Design vs. Native Apps

| Approach | Technology & Architecture | Pros & Cons |
|---|---|---|

| | | |
|---|---|---|
| **Responsive Web Design (RWD)** | Fluid grid layouts, flexible CSS media queries. Single codebase for all devices. | **Pros:** Cost-effective, universal reach, single maintenance URL.<br>**Cons:** Limited access to native phone hardware, requires persistent connection. |
| **Native Mobile App** | Compiled platform code (Swift for iOS, Kotlin for Android). | **Pros:** Peak performance, full hardware integration (GPS, camera, push notifications, biometrics), offline capabilities.<br>**Cons:** High development cost, requires app store approval and dual codebases. |

# 04 E-Commerce Security and Payment Systems (Chapter 5)

## 4.1 The 6 Dimensions of E-Commerce Security

| Security Dimension | Customer's Perspective | Merchant's Perspective |
|---|---|---|
| **Integrity** | Has the information I transmitted or received been altered? | Has data on the site or in the database been modified by unauthorized parties? |
| **Non-repudiation** | Can a merchant deny having received my payment? | Can a customer order goods and falsely deny having placed the order? |
| **Authenticity** | Am I dealing with the real merchant and authentic website? | What is the real identity of the customer making this purchase? |
| **Confidentiality** | Can unauthorized third parties view my messages or credit card details? | Are corporate proprietary trade secrets and consumer data protected from snooping? |
| **Privacy** | Can I control the use and distribution of my personal profile data? | Is personal information collected responsibly and compliant with privacy laws? |
| **Availability** | Can I access the e-commerce website whenever I need to shop? | Is the server infrastructure resilient against crashes, outages, and DoS attacks? |

## 4.2 Major Security Threats

- **Malware:** Viruses (require host file execution), Worms (self-replicating autonomous programs), Trojan Horses (disguised malicious programs), Ransomware (extortion via data encryption).
- **Botnets:** Networks of compromised zombie computers used to launch massive Distributed Denial of Service (DDoS) attacks.
- **Phishing & Spoofing:** Deceptive emails and counterfeit clone websites designed to steal credentials. Pharming redirects domain traffic directly.
- **Hacking:** White hats (ethical security researchers), Black hats (malicious criminals), Grey hats (unauthorized researchers without criminal intent).
- **Sniffing & Man-in-the-Middle (MitM):** Eavesdropping on unencrypted network traffic or intercepting communication streams.

## 4.3 Technology Solutions: Cryptography & PKI

| Technology | Mechanism | Primary Application |
|---|---|---|
| **Symmetric Encryption** | Single shared secret key used for both encryption and decryption (AES, DES). Fast. | Bulk payload data encryption. *Key distribution is the major challenge.* |

| | | |
|---|---|---|
| **Asymmetric Encryption (Public Key)** | Two mathematically linked keys: Public Key (shared freely for encryption) and Private Key (secret, for decryption). | Secure key exchange, digital signatures (RSA, ECC). Solves key distribution. |
| **Digital Signature** | Message digest hashed and encrypted with the sender's Private Key. | Guarantees **Integrity** and **Non-repudiation**. Verified using sender's public key. |
| **Digital Certificate & CA** | Digital document issued by a trusted Certificate Authority (CA) binding a public key to an entity's verified identity. | Guarantees **Authenticity**, enabling HTTPS/SSL/TLS communication. |
| **SSL/TLS & HTTPS** | Combines asymmetric handshake for session key exchange and symmetric encryption for data stream. | Secures client-server communication channels over the open Web. |

### 4.4 Online Payment Systems & Protocols

- **Credit Card Transaction Flow:** Consumer -> Merchant Gateway -> Acquiring Bank -> Card Brand Network (Visa/MC) -> Issuing Bank (authorizes fund transfer).
- **Digital Wallets (Apple Pay, PayPal, Google Pay):** Store payment details securely. Modern wallets use **Tokenization**: replacing actual 16-digit card numbers with dynamic, one-time encrypted tokens.
- **3D Secure (Verified by Visa, Mastercard Identity Check):** Adds multi-factor authentication (OTP/biometrics) during checkout, shifting chargeback fraud liability to the card issuer.
- **PCI-DSS Compliance:** Mandatory security standard governing all organizations handling cardholder data to protect against breaches.